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DESCRIBE THE TYPES OF COMPUTER SYSTEMS AND ASSOCIATED HARDWARE CONFIGURATIONS

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SECTION A: FORMATIVE ASSESSMENT

QUESTION 1

1. MAINFRAME COMPUTERS are data processing system kept in separate rooms because of their size. It has 1 to 16 CPU's, memory ranges from 128 Mb over 8 Gigabyte on line RAM, it's processing power ranges from 80 over 550 Mips, it has often different cabinets for storage, I/O and RAM, there are separate processes for tsk management, program management, job management, serialization, catalogues, inter address space and communication.
2. MIDRANGE COMPUTERS are computers that stand in the middle of mainframes and microcomputers. The computer or server supports a standard interface used in local area network
3. MICROCOMPUTERS get their name from the word micro which means small, that is their overall size and they use microprocessors as the CPU to process and store data, most computers are too big to carry around easily and practically like desktops , but other like laptops are portable.

QUESTION 2

1. Processing speed is how fast the CPU executes instructions , Mainframes have very high speed for batch jobs
2. Throughput : the amount of work done in a given time. Mainframes handle huge transaction volumes.
3. Capacity: RAM and storage size. A PC might have 16GB RAM, a mainframe has TBs
4. Reliability and availability: Mainframes run 24/7 with little downtime. PCs may need restarts.
5. Scalability: Ability to add more users or resources. Mid-range systems are easier to scale than PCs.

QUESTION 3

The similarities of personal computers and mainframes are:

1. Need stable power
2. Need security to prevent unauthorised access
3. Need clean , dust free environment

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Differences

	Personal computer	Mainframe
Space	Normal desk or office	Special raised floor computer room
Cooling	Standard air conditioning	Industrial cooling and humidity control
power	Normal wall socket	Dedicated power lines, large UPS , generator backup
Access	Single user, no special security	Restricted access, fire suppression systems
Cost of environment	low	Very high

QUESTION 4

Applications are grouped by purpose so it's easier to design , maintain and choose the right software.

1. Scientific applications is used in sciences which are specifically designed for the field in which they are used. This hardware requires very high specifications and involves calculations.
2. Business applications include spreadsheets, database, presentation, word processing, and project management applications ;it manages data and transactions for organizations.
3. Home applications , this application mostly involves entertainment
4. Educational applications are used to teach a wide range of skills ,you can access information

QUESTION 5

1. Scientific applications is an application that performs complex scientific calculations , it takes usually more than one computer to run it, needs high computation power and accuracy , and can be slower but must be precise.
2. Business applications , most businesses rely on users' being connected to each other via a network. The business computers need to be networked PCs to fully benefit from business application, needs high throughput , data integrity and availability and must handle many users and large database without errors.
3. Home application need less processing power because they are used to perform basic functions
4. Educational application do not need much processing power or speed because they display information